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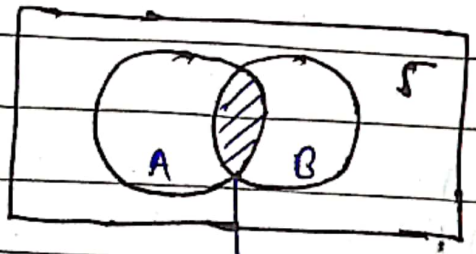
(#) Types of events :

- Joint event
- Disjoint
- Dependent
- Independent

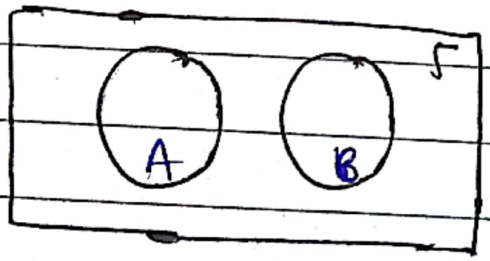
Joint event :

(also called Non Disjoint / Non-mutually Exclusion)



common outcomes

Disjoint :



→ Disjoint also called "mutually Exclusive" (do not occur at same time).

Joint event :

$P(A \text{ and } B) = P(A) * P(B)$

$P(A \text{ or } B) = P(A) + P(B) - P(A \& B)$

Distant :

$P(A \text{ and } B) = 0$

$P(A \text{ or } B) = P(A) + P(B)$

Dependent events :

D.E is like when one event is influence by another.

→ $A \rightarrow B$

$A \Rightarrow$ Drawing a red card

$B \Rightarrow$ Drawing a black card

$\left(A = \frac{26}{52}\right)$, $\left(B = \frac{26}{51}\right)$

→ B is influence by A. in this example

→ $\boxed{P(A \text{ and } B) = P(A) * P(B/A)}$

$P(B/A)$ = probability of B given A

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(ii) Independent Day :-

When event A does not affect another one

ie: A = Tossing a coin → {H, T}
B = Tossing a 2nd coin → {H, T}

$$\boxed{P(A \& B) = P(A) \cdot P(B)}$$

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